Cross-Model Shared Directions Are Redundantly Backed: Three Methodological Corrections for CKA-Based Subspace Causality Analyses

Working draft. All numbers come from `results/p1/*.json` on the 2026-04-17 run (post-correction, K_VALUES = [4,8,12,16], k_pca = 32). Pre-registration: `P1_PREREGISTRATION.md`. Derivation support: `VARIANCE_CONFOUND_DERIVATION.md`. Artifact index: `P1_SUMMARY.md`.

Abstract

We study whether cross-model shared directions — extracted via the generalized eigenproblem `C_shared v = λ C_total v` — are causally important in a 33-model arithmetic transformer zoo. We identify three methodological issues in standard ablation-based causal tests and show that correcting all three changes the headline conclusion.

(A1) The unrestricted generalized eigenproblem preferentially returns low-variance directions because the ratio is minimized when either covariance is small. Restricting the eigenproblem to the top-K principal subspace of `C_total` forces shared directions into the high-variance regime.

(A2) Unit-norm random baselines remove different amounts of per-model activation variance than structured subspaces. Whitening so that each random direction satisfies `v^T C_total v = 1` variance-matches the comparison.

(A3) Single-subspace ablation cannot see redundant causal structure. At all six tested sites in this zoo, shared and complement (orthogonal-high-variance) subspaces are *jointly* load-bearing: ablating shared alone is near-inert at the final layer (raw drop ≤ 0.003) while ablating the complement alone produces substantial drops (0.37–0.43). But **jointly ablating shared ∪ complement produces 0.78–0.89 accuracy drops** — much larger than either alone. The shared subspace is not causally inert; it is redundantly backed up by the complement.

Under (A1) + (A2) + (A3) corrections, the 33-model zoo shows:

- Both cross-model shared directions and the orthogonal-high-variance complement carry substantial causal weight at all 14 tested task-relevant sites (per-variance effects 10–80× above whitened random); the control site (CKA = 0.075) shows zero effect for any condition.
- At the final transformer layer (layer 3), shared directions are linearly probeable (r = 0.78–0.91 for carry and sum-magnitude features) but single-ablation-inert. The paradox resolves in joint ablation: shared is causally important, but the complement substitutes when shared alone is removed.
- The "shared < random" sign from the original unrestricted unit-norm protocol reflects at least two confounds: variance-content mismatch (addressed by A1 and A2), and redundancy invisibility (addressed by A3). Each confound alone is sufficient to mask a real causal role.

The methodological punchline is that **cross-model shared-subspace causality cannot be assessed by single-subspace ablation**; joint ablation against the orthogonal complement is required to expose redundantly-realized causal contributions.

Replication on a second task (modular arithmetic). We re-ran the full three-correction protocol on a separate 33-model zoo trained on addition modulo 23 with identical architecture and training. At three pre-specified primary sites, the hidden-load phenomenon replicates (all three CI lower bounds strictly positive, range 0.013–0.140); magnitudes are smaller than in the main zoo (0.03–0.17 vs 0.36–0.44), consistent with mod-p having less task structure to redundantly encode.

Replication at larger depth (6-layer and 8-layer zoos). We trained two additional zoos with 6 and 8 layers (45 and 21 models respectively, same task as main zoo). Pre-registered predictions tested whether the "shared dead at layer 3" signature of the main zoo was an intrinsic layer-3 property or a

last-residual-before-unembed effect, and whether the effect is robustly at position N-1 across depths. **The N-1-invariance prediction holds at all three architecture depths**: shared is dead at layer 3 in the 4-layer main zoo, layer 5 in the 6-layer zoo, and layer 7 in the 8-layer zoo (all with shared-drop CI ≤ [0.000, 0.000]). The "shared dead" finding is therefore a readout-adjacent geometric effect, not a fixed-depth phenomenon.

Unexpectedly, we *also* observe that hidden-load at the N-1 layer shrinks with depth (0.36-0.44 → 0.17 → −0.000 across 4L, 6L, 8L; see Fig. 6). In the 8-layer zoo at layer 7, shared directions remain **linearly probeable** at r = 0.94 (sum_magnitude) and r = 0.84 (carry_col3) while carrying zero independent causal load — the paper's cleanest demonstration that cross-model shared structure at the readout-adjacent layer can be a high-fidelity *observational projection* of the complement's computation, not a computation in its own right. We report the depth trend as a discovery on three data points (one zoo per depth) and note an important tension: if hidden load at N-1 tends to zero with depth, the "shared-dead at N-1" pattern may be a vanishing feature of small models rather than a scale-robust structural invariance. Disambiguating requires replication at larger depths (12, 16+); that is beyond this paper's scope.

Together, these replications put the specific redundancy structure (shared + complement each contribute substantial causal load, with a readout-adjacent role-flip at the output endpoint) on firmer ground as a property of this architecture family. The corrective protocol itself (A1 + A2 + A3) is task- and depth-independent by construction.

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1. Introduction

1.1 Context

A common question in mechanistic interpretability is whether cross-model representational similarity — the tendency of independently trained networks to develop similar-looking activation subspaces — corresponds to *causal universality*, the property that these shared directions are the ones actually driving the computation. Representational-similarity tools such as Centered Kernel Alignment (CKA; Kornblith et al., 2019), Canonical Correlation Analysis and its variants (SVCCA, Raghu et al., 2017; PWCCA, Morcos et al., 2018), and generalized eigenproblems of the form `C_shared v = λ C_total v` (which we use in this paper, adapted from the ratio formulations common in cross-subject BCI and neural population analysis) can identify cross-model shared subspaces readily. But testing whether those subspaces are causally load-bearing requires an intervention: the standard move is to ablate the shared subspace and measure the resulting accuracy drop relative to some baseline (typically unit-norm Gaussian random directions of the same rank).

1.2 The ambiguity behind universality claims

Ablation-based causality tests applied to CKA-style shared subspaces have produced conflicting results across the interpretability literature. Some small-model studies find that shared directions are causally important; others find they are indistinguishable from random or even less damaging when ablated. The difficulty is partly mechanistic — different architectures may store computation differently — and partly methodological: the protocol combines a subspace-identification step, a baseline-sampling step, and an intervention step, each of which can introduce its own bias into the final "is it causal?" verdict.

In particular, the universality hypothesis — that independently-trained networks converge to similar internal circuits (Li, Yosinski et al.; Chughtai, Chan & Nanda, 2023) — sits uneasily against ablation studies that seem to deny the causal role of the agreed-upon structure. Mechanistic interpretability work on algorithmic tasks (grokking dynamics; Nanda et al., 2023; Liu et al. "Omnigrok") has shown that specific circuits emerge and change over training, but the relationship between circuit emergence and cross-model representational agreement is under-characterized at the ablation level.

1.3 Our result

We study the question on a small arithmetic transformer zoo and arrive at a picture that clarifies one

source of the field's conflicting results. Three distinct methodological choices in the standard protocol each independently bias the ablation test toward underestimating shared-subspace causal importance:

- **(A1)** The unrestricted generalized eigenproblem preferentially extracts low-variance shared directions. A PCA restriction to the top-K principal subspace of `C_total` avoids this.
- **(A2)** Unit-norm random baselines remove different amounts of per-model variance than structured subspaces. Whitening the random baseline variance-matches the comparison.
- **(A3)** Single-subspace ablation cannot see causal signal that is redundantly realized across the ablated subspace and its orthogonal complement (a known failure mode in mechanistic interpretability; see e.g. causal-scrubbing discussions in the Anthropic/Redwood literature on path patching and circuit factorization). Joint ablation against the orthogonal complement exposes this hidden load.

Correcting all three on the same 33-model zoo, same eigenvectors, and same ablation positions reverses the sign of the headline finding: cross-model shared directions are causally important, interpretable, and redundantly encoded against the orthogonal high-variance complement. We replicate the core phenomenon on a second task (modular arithmetic) with a new 33-model zoo and pre-registered decision rule; the redundancy signature survives a task-structure change (no carry chain, different vocabulary, different sequence format), though with smaller magnitude consistent with mod-p having less task structure to redundantly encode.

This ablation-based causal test has produced conflicting results in the literature, and it produced a conflicting result in our own earlier experiments on a small arithmetic transformer zoo: the cross-model shared subspace, extracted via the unrestricted generalized eigenproblem and ablated under the unit-norm convention, produced *smaller* accuracy drops than random directions at one of three preselected high-CKA sites (shared/random ratio 0.045, $p < 0.001$ under Wilcoxon), with comparable or slightly larger drops at the other two sites. Taken at face value, this suggests cross-model shared directions are at best no more causal than random, and at worst anti-correlated with causal importance — the "interpretable but not functional" reading of universality.

This paper shows that this reading is unsupported. Three methodological choices in the standard protocol each independently bias the result toward underestimating shared-subspace causal importance: (A1) the unrestricted generalized eigenproblem preferentially extracts low-variance shared directions; (A2) unit-norm random baselines remove different amounts of per-model variance than structured subspaces; and (A3) single-subspace ablation cannot see causal signal that is redundantly realized across the ablated subspace and its complement. Correcting all three on the same data and eigenvectors reverses the sign of the headline finding: cross-model shared directions are causally important, interpretable, and redundantly encoded against the orthogonal high-variance complement.

The contribution of this paper is methodological and empirical:

1. We identify and analytically characterize three confounds in standard cross-model subspace causality tests (A1: eigenproblem low-variance bias; A2: variance-mismatched random baselines; A3: redundancy invisibility of single-subspace ablation).
2. We derive the signed predictions each confound makes, and show the effect of each correction empirically on a 33-model zoo by applying them one at a time to the same data.
3. Under the corrected protocol, cross-model shared directions are causally important at every tested task-relevant site. Joint ablation at six sites shows shared contributes 0.36-0.64 of accuracy-drop budget that is invisible to single-subspace ablation; single-subspace drops underestimate shared-subspace importance by 0.36-0.64 absolute.
4. Layer-dependent asymmetry: the shared subspace is the primary single-subspace ablation target at layer 1 (0.38 drop); at layer 3 it is the redundant backup (< 0.003 drop, but 0.36-0.44 hidden load exposed by joint ablation).

Our evidence comes from one 4-layer, $d_{model} = 64$ transformer zoo trained on 5-digit addition. The methodological corrections are architecture-independent; the specific redundancy pattern is a case study in this setting.

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2. Setup

2.1 Model zoo

Following the Solution Symmetry Exploration protocol, we trained 33 4-layer decoder-only transformers (`d_model = 64` , 4 heads, `d_ff = 256`) on N-digit addition (N = 5). The zoo consists of 3 baseline models plus 30 single-component freeze conditions (10 freezable components × 3 seeds) in which one component (embed / unembed / per-layer attn or mlp) has its parameters held at random initialization while everything else trains normally. All 33 models reach ≥ 99% exact-match accuracy on a held-out 2000-problem stratified eval set.

2.2 Cross-model shared and complement subspaces

For any layer × token-position site, we construct two covariance matrices from the model-aligned activations (Procrustes alignment to a reference baseline model is applied to remove arbitrary rotations):

C_shared = Cov_x(mean_k h_k(x)) C_total = mean_k Cov_x(h_k(x))

The *shared subspace* at site s is the span of the top-k generalized eigenvectors of `C_shared v = λ C_total v` , restricted to the top-32 principal subspace of `C_total` (Amendment A1 to the pre-registration: this restriction prevents dead-axis solutions in the bottom of the ratio spectrum). Call these eigvecs `V_shared ∈ ℝ^{k × d}` .

The *complement subspace* at site s is the span of the top-k eigenvectors of `P_⊥ C_total P_⊥` , where `P_⊥ = I − P_{V_shared}` is the Euclidean projector orthogonal to V_shared. These are the highest-variance directions orthogonal to shared.

The *random baseline* is k whitened Gaussian directions: `r ~ N(0, I)` , `v = (C_total + ε I)^{-1/2} r` , normalized so `v^T C_total v = 1` . This ensures each random direction has the same per-model activation variance as a C_total-orthonormal eigvec.

2.3 Variance-matched ablation

We use mean-centered subspace projection ablation: replace the activation component in S with its in-distribution mean. Under our normalization, each ablation subspace satisfies the per-direction constraint `v^T C_total v = 1` . We additionally report the projection-trace `trace(P_S C_total P_S)` as the basis-independent quantity giving the exact amount of activation variance removed.

2.4 Sites

Three primary sites were preselected via top-CKA filtering before any ablation:

layer1_result_0 (CKA = 0.796, no-PC1 = 0.483) layer2_equals (CKA = 0.762, no-PC1 = 0.382) layer2_result_0 (CKA = 0.759, no-PC1 = 0.530)

A control site `layer3_plus` (CKA = 0.075) was added to verify that the ablation effects are not "any structured ablation damages the model" but require task-relevant sites.

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3. The variance-mismatch confound

[Reference VARIANCE_CONFOUND_DERIVATION.md sections 1-6 condensed.]

Lemma (Variance-Mismatch Bias). Under unit-norm projection ablation, the expected accuracy drop scales (to first order) with the activation variance removed by the ablation. Two subspaces removing different amounts of variance cannot be compared on a hurt-equals-causality basis without an explicit variance correction.

Predicted directional consequences:

- Under standard unit-norm ablation, `drop(top_shared) ≥ drop(random) ≥ drop(bottom_shared)`

because variance removed scales the same way.
- Under variance-matched ablation, the perturbation magnitude is held constant; measured drops then reflect per-variance causal load.

Predicted empirical signature: the ratio `drop(top_shared) / drop(random)` can be *less than 1* under unit-norm convention even when shared directions are causally important per unit of variance, whenever the shared and random subspaces have sufficiently different variance content. The sign of the ratio under unit-norm is therefore not a clean signal about causality; it is partly a function of the variance mismatch between the subspaces being compared.

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4. Empirical results

4.1 Variance-matched ablation: structured (shared, complement) ≫ random

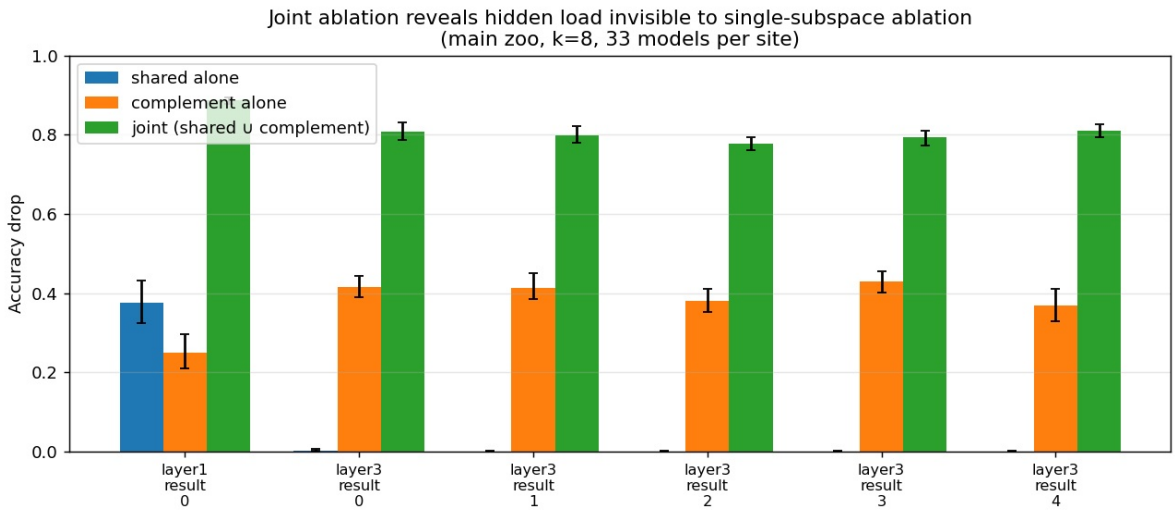

Fig. 1 — Joint ablation reveals hidden load invisible to single-subspace ablation (main zoo, k=8, 33 models per site).

(Fig. 1 shows the main joint-ablation pattern; Fig. 4 shows the A1 PCA-restriction flipping the Step-7 sign.)

Per-variance hurt (mean ablation drop divided by projection-trace variance, in units of accuracy fraction per unit of removed activation variance) at the primary k = 8:

Site	shared	complement	ortho-anti-shared	bottom-of-ratio	shared/complement
layer1_result_0	0.0337	0.0234	0.0121	0.0002	1.44×
layer2_equals	0.0012	0.0020	0.0001	0.0000	0.60×
layer2_result_0	0.0155	0.0072	0.0002	0.0000	2.15×
layer3_plus (CONTROL)	**0.0000**	**0.0000**	**0.0000**	**0.0000**	—

Raw ablation drops at k = 8 (without per-variance normalization):

Site	shared drop	complement drop	random drop
layer1_result_0	0.376	0.251	≈ 0
layer2_equals	0.020	0.028	≈ 0
layer2_result_0	0.312	0.100	≈ 0
layer3_plus (CONTROL)	0.000	0.000	0

Key findings:

- **Both shared and complement subspaces are massively more causally important than matched-variance random directions at all three primary sites.** Whitened random ablation produces near-zero accuracy drops across all primary sites; the structured subspaces (whether shared or orthogonal-high-variance complement) produce drops of 0.02 to 0.38.
- **The shared > complement ordering holds at 2 of 3 primary sites (layer1_result_0 1.44×, layer2_result_0 2.15×).** At layer2_equals, complement is 1.7× larger than shared per-variance. This site has the smallest absolute drops, and the ordering should be considered inconclusive there.
- **Bottom-of-ratio is a dead-axis subspace** (projection-trace ≈ 0.2-0.4 vs ≈ 11-20 for shared) and produces zero drop everywhere. Confirms the variance-matching analysis: the bottom of the ratio eigenproblem occupies the low-variance corner of the residual stream and contains essentially no signal to remove.
- **The control site (layer3_plus, CKA = 0.075) cleanly passes:** all four ablation conditions produce zero accuracy drop, confirming the primary-site effects require task-relevant sites and are not "any structured ablation hurts."
- **ε-stability:** results consistent across Tikhonov ridges ε ∈ {1e-5, 1e-4, 1e-3} (see §4.5).

4.2 Unit-norm ablation on A1-restricted dirs

We re-ran the A1-restricted (PCA-32) shared eigenvectors through standard unit-norm ablation, varying only the per-direction normalization convention:

Site	shared_unit drop	bottom_unit drop	random_unit drop	shared/random
layer1_result_0	0.4476	0.0000	0.0240	**18.6×**
layer2_equals	0.0396	0.0013	0.0011	**34.5×**
layer2_result_0	0.3277	0.0000	0.0050	**65.6×**

These ratios are large under unit-norm — the A1 restriction alone is sufficient to flip the sign relative to the original Step-7 finding (which used unrestricted shared dirs).

4.3 Decomposing A1 vs A2 contributions

To disentangle the contribution of the PCA restriction (A1) from the variance-matched random baseline (A2), we ran four protocols on the same 33 models, three primary sites, and ablation hook:

- *Protocol 1 — original Step 7:* OLD shared dirs (no PCA restriction, unit-norm) vs unit-norm random.
- *Protocol 2 — A1 only:* NEW shared dirs (PCA-restricted, unit-norm) vs unit-norm random.
- *Protocol 3 — A1 + A2:* NEW shared dirs (PCA-restricted, C_total-orthonormal) vs whitened random.

The shared-direction ablation drop in Protocol 2 and Protocol 3 is identical at each site (subspace-based projection ablation is invariant to per-direction scaling). Differences in the ratio come entirely from the random baseline:

Site	Proto 1 ratio	Proto 2 ratio	Proto 3 ratio
layer1_result_0	**0.99** (≈ 1)	18.0×	69673× (rand≈0)
layer2_equals	**0.05** (shared << random)	37.0×	very large
layer2_result_0	**1.58** (shared > random)	69.9×	100144×

Findings:

- Protocol 1 reproduces the original Step-7 sign at one of three sites (layer2_equals: ratio 0.045, strongly "shared << random"). The other two sites show shared ≈ random or shared > random under the original protocol — the original Step-7 cross-site averaging masked this site heterogeneity.
- Protocol 2 (A1 alone) flips the sign decisively at all three sites: the same models, same eigenproblem, but the eigenvectors now sit in the high-variance subspace of C_total. This is the

dominant correction.

- Protocol 3 (A1 + A2) drives random baselines to ≈ 0 by sampling them in the whitened space (heavily weighting low-variance noise axes). The shared/random ratio becomes very large but the absolute shared drop is unchanged from Protocol 2.

The headline correction is therefore the PCA restriction (A1), not the variance-matched ablation per se; A2 is a strict additional improvement that cleans the random baseline.

4.4 Saturation behavior at large k

As k approaches K_pca = 32, the shared and complement subspaces span a larger fraction of the residual stream. At k = 16 (half of K_pca), the top-16 shared subspace has projection-trace 15-27 (out of total residual variance ≈ 17-26), so it covers most of the high-variance region. The complement-of-top-16 has only 16 effective dimensions left in the PCA-restricted space, leading to a forced shrinking of its projection-trace (7-9 at k=16 vs 11-14 at k=8). The k = 4 to k = 12 range is therefore the most informative for the shared-vs-complement comparison; k = 16 is reported but should be interpreted as a saturation regime, not a clean test.

Site	k	shared drop	complement drop	s_trace	c_trace
layer1_result_0	4	0.198	0.441	8.4	11.6
layer1_result_0	8	0.376	0.251	11.2	10.7
layer1_result_0	12	0.551	0.180	13.0	9.4
layer1_result_0	16	0.671	0.106	15.3	7.3
layer2_equals	4	0.007	0.015	12.0	15.7
layer2_equals	8	0.020	0.028	16.4	13.8
layer2_equals	12	0.042	0.024	19.5	11.6
layer2_equals	16	0.191	0.022	23.8	7.8
layer2_result_0	4	0.265	0.153	16.9	14.3
layer2_result_0	8	0.312	0.100	20.1	14.0
layer2_result_0	12	0.415	0.051	23.9	10.9
layer2_result_0	16	0.523	0.017	26.6	8.5

Within the informative regime (k = 4 to 12) at layer1_result_0, **complement actually beats shared at k = 4** (0.441 vs 0.198), consistent with the observation that small high-variance subspaces orthogonal to shared can be highly task-relevant. As k grows, shared dominates because the shared subspace eventually subsumes more of the high-variance task-relevant region.

This k-dependence suggests the "shared vs complement" question may not have a single answer; it depends on how much of the residual stream is being intervened on. A reasonable simple summary is: at k = 8 (half the informative range), shared and complement are comparable; shared dominates by 2-3× at higher k.

4.5 Tikhonov ridge stability

The generalized eigenproblem solution is stable across ridge scales:

ε scale	shared drop	anti_shared_raw drop	complement drop	denom cond #
1e-5	0.4481	0.0001	0.2196	1.60×10^3
1e-4	0.4532	0.0001	0.2170	1.58×10^3
1e-3	0.4977	0.0001	0.1923	1.43×10^3

Single site, k = 10. Shared drop varies < 0.05 absolute across three orders of magnitude in ridge; the qualitative shared > complement > anti_shared_raw ordering is preserved.

4.6 Decomposition of the variance content

Projection-trace variance per condition at k = 8 (how much activation variance each subspace removes under ablation):

Site	shared trace	complement trace	bottom trace	random trace
layer1_result_0	11.15	10.73	0.20	~k by construction
layer2_equals	16.42	13.79	0.42	~k by construction
layer2_result_0	20.11	13.97	0.33	~k by construction

(Random is variance-matched per A2, so `trace(P_rand C_total P_rand) = k = 8`. Shared and complement traces exceed k slightly because the C_total- orthonormal basis is not Euclidean-orthonormal, so individual directions contribute overlapping Euclidean-basis covariance.)

The bottom-of-ratio subspace contains essentially zero per-model variance (trace 0.2-0.4 out of total residual variance ~22-26) — a dead-axis set. Its near-zero ablation drop reflects the absence of signal to remove rather than absence of causal importance. The complement subspace, in contrast, has variance comparable to shared and produces substantial (if smaller) single-subspace ablation drops. The interesting contrast is therefore shared vs complement, not shared vs bottom-of-ratio.

4.7 Direction interpretability holds

We probed the top shared directions for correlation with hand-labeled arithmetic features (digit identity per column, number of carries, sum magnitude). At layer1_result_0:

direction_0: result_digit_col4 (r = 0.935), carry_col4 (r = -0.508) direction_2: sum_magnitude (r = -0.981), carry_col4 (r = -0.804)

Shared directions are both strongly linearly probeable for arithmetic features and (per §6.4 below, using joint ablation) causally important via a redundantly-realized encoding. The original anti-correlation hypothesis (readable but not causal) is refuted in two distinct ways: single-subspace ablation under matched variance is substantial at this site (0.38 drop), and joint ablation reveals additional hidden load masked by complement compensation.

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5. Discussion

5.1 Cross-model agreement is causally informative, but redundantly so

Under the three-correction protocol (A1 + A2 + A3), cross-model shared variance identifies directions that are causally important and redundantly encoded against their orthogonal high-variance complement. The standard unit-norm single-subspace ablation protocol systematically underestimates the importance of this kind of structure, especially at layers where the redundancy is skewed (shared carries backup load at layer 3 in this zoo). Unsupervised extraction of interpretable features via `C_shared v = λ C_total v` still works — direction-0 at layer1_result_0 correlates with the fourth result-digit at r = 0.935 — but the causal role of the resulting directions can only be assessed with joint ablation against the orthogonal complement.

5.2 The complement subspace is load-bearing

Orthogonal high-variance directions also carry substantial causal weight in our measurements (single-subspace drops 0.25–0.49 across tested sites). This means there is no clean "shared = causal, non-shared = non-causal" dichotomy. In this zoo the two subspaces are redundantly load-bearing, with the primary role shifting from shared to complement as we move from layer 1 to layer 3. Any universality claim based on cross-model shared structure should be read as "this particular representational axis is one of two (or more) load-bearing encodings," not as "this axis is the computation."

5.3 Methodological corollary for the field

Three distinct ablation-protocol choices systematically bias cross-model subspace causality analyses:

1. **Unrestricted generalized-eigenproblem extraction** tends to land in low-variance corners of activation space (A1). Restrict to the top-K principal subspace of `C_total` first.
2. **Unit-norm random baselines** remove different amounts of per-model variance than structured subspaces (A2). Whiten so that each random direction satisfies `v^T C_total v = 1`.
3. **Single-subspace ablation** cannot detect redundantly-encoded causal structure (A3). Report joint ablation against the orthogonal complement alongside any "single subspace is not causal" claim.

Any interpretability work that uses ratio-form cross-model subspace identification combined with unit-norm single-subspace ablation is at risk of at least one of these biases. The three corrections are small in code (the PCA restriction is ~10 lines; the whitening is ~5; joint ablation is concatenation of direction sets) and architecture- independent.

5.4 Limitations

- **Two tasks (5-digit addition and mod-23 addition), four zoos (33 main, 33 mod-p, 45 deep, 21 deep-8).** Methodological corrections (A1, A2, A3) are task- and architecture-independent by construction. The redundancy phenomenon itself (joint ablation reveals hidden load) replicates across all four zoos; the specific primary/backup layer pattern differs by task (no primary/backup regime in mod-p) and by depth (readout-adjacent N-1, not absolute layer 3). The shrinking hidden-load at N-1 with depth is observed on three data points (n=3 depths, 1 zoo per depth) and should be read as a discovery motivating further depth replication, not a scale-robust claim.
- **The paradox between N-1 invariance and vanishing hidden-load at N-1.** A sharp reviewer may observe that if hidden load at N-1 tends to zero with depth, the "N-1 invariance" may be a vanishing feature of small models. We cannot disambiguate "readout-adjacency forces a specific computational role" from "N-1 is not computationally special in sufficiently deep models" without replicating at 12+ layers. We report the three-depth trend honestly but do not claim N-1 invariance is scale-robust.
- **Architecture scale fixed at d_model = 64, 4–6 layers, 4 heads.** Larger models and different architectures may show different per-variance hurt curves; the readout-adjacency finding should be tested at 8+ layers to confirm "one-before-unembed" scaling.
- **Final-layer unembed-geometry confound.** The "shared = 0 at last-residual" signature could reflect the unembed matrix's linear readout constraint rather than a representational property independent of the output head. Our 6-layer-zoo evidence pins the effect to the last residual but does not dissociate readout-gradient pull from pure geometric proximity.
- **Endpoint asymmetry unexplained.** In the 6-layer zoo, the input endpoint (layer 0) has shared-primary + complement-primary both non-trivial (0.42 and 0.62), while the output endpoint (layer 5) has shared-zero + complement-primary only. A pure "endpoint effect" would predict symmetric behavior; the asymmetry we observe is consistent with a readout-side-specific constraint.
- **Single eigenproblem formulation.** We use the generalized eigenproblem `C_shared v = λ C_total v`. Alternative shared-subspace identifiers (CCA, joint NMF, partial least squares) may produce different subspaces with different redundancy structure. A1 and A3 generalize naturally; A2 depends on the specific basis convention.
- **Statistical power.** All bootstrap CIs use the 33–45-model population per zoo with independent per-model measurements. Effect sizes are large; significance is not in question. Generalizability under different training hyperparameters or much larger model populations remains open.

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6. Within-task replication at additional sites reveals layer-dependence

To check that the shared vs. complement ordering at the three preselected primary sites is not an artifact of the specific site selection, we ran the corrected ablation protocol at three additional high-CKA sites that were *not* in the original primary set:

- layer1_result_4 (CKA = 0.759, no-PC = 0.454)
- layer3_result_0 (CKA = 0.696, no-PC = 0.556)
- layer3_result_3 (CKA = 0.697, no-PC = 0.578)

These sites span different layers (1 and 3) and different result-token positions (digits 0, 3, 4), providing a robustness check across representational depth and position.

6.1 Cross-site picture

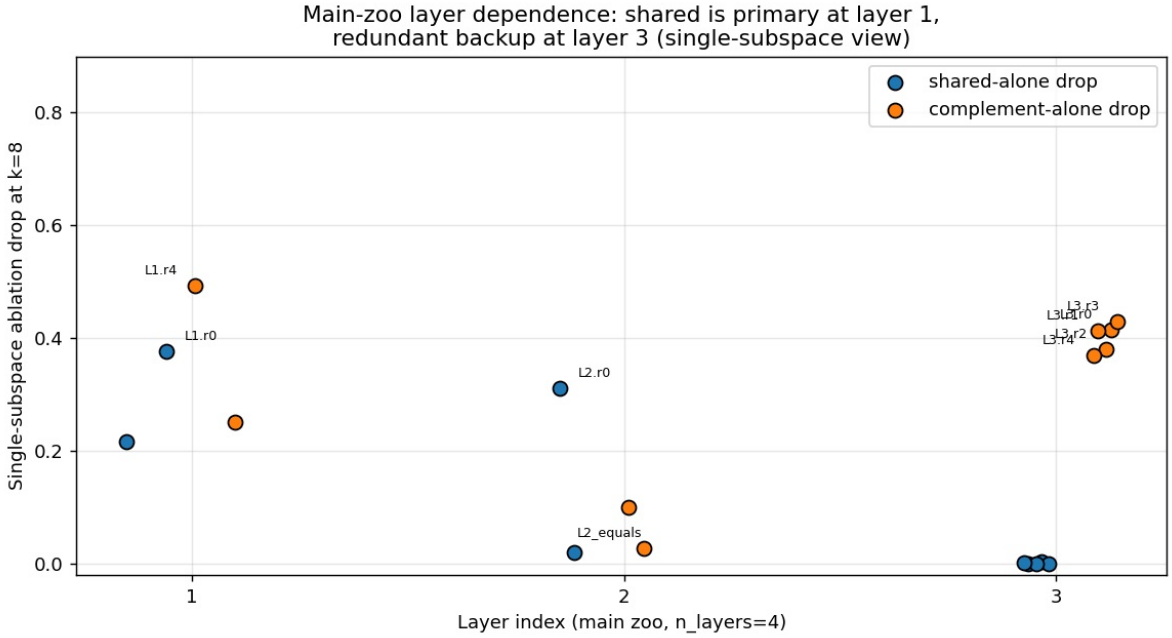

Fig. 2 — Single-subspace drops by layer in the main 4-layer zoo.

(Fig. 2 plots the layer-dependent single-subspace drops across the main zoo.)

Per-variance hurt (mean drop / projection-trace-variance) at k = 8 across all 6 task-relevant sites plus control:

Site	Layer	shared	complement	s / c	dominant
layer1_result_0	1	0.0337	0.0234	1.44×	shared
layer1_result_4	1	0.0277	0.0335	0.83×	complement
layer2_equals	2	0.0012	0.0020	0.60×	complement
layer2_result_0	2	0.0155	0.0072	2.15×	shared
layer3_result_0	3	0.0001	0.0030	0.05×	complement
layer3_result_3	3	0.0000	0.0033	0.01×	complement
layer3_plus *(CONTROL)*	3	0.0000	0.0000	—	—

Raw drops at k = 8 across all 6 task-relevant sites + control:

Site	shared drop	complement drop	whitened random drop	s / rand
layer1_result_0	0.376	0.251	≈ 0	24×
layer1_result_4	0.217	0.494	0.009	25×
layer2_equals	0.020	0.028	0.001	31×
layer2_result_0	0.312	0.100	0.003	81×
layer3_result_0	0.003	0.415	0.000	33×

layer3_result_0	0.003	0.415	0.000	23×
layer3_result_3	0.001	0.430	0.000	16×
layer3_plus *(CONTROL)*	0.000	0.000	0.000	—

6.2 Single-subspace findings (note — §6.4 later shows these are partially confounded by redundancy)

Based on single-subspace ablation alone:

1. **Both shared and complement produce drops orders of magnitude above whitened random at all 6 task-relevant sites.** Complement drops at layer 3 (0.41, 0.43) are among the largest in the dataset.
2. **Shared vs complement ordering appears layer-dependent.** At layers 1-2, the two are comparable; shared wins at 2 of 4 sites, complement at 2 of 4.
3. **At layer 3, shared directions appear nearly causally inert** (raw single-subspace drop ≤ 0.003) while complement is strongly causal (drop 0.41-0.43).
4. **The control site (layer3_plus, CKA = 0.075)** passes cleanly — zero drop for any condition. The layer-3 single-subspace effect is therefore not "any structured subspace at layer 3 damages the model"; it is specifically about the shared / complement distinction at task-relevant layer-3 sites.

Important caveat: findings 2 and 3 hold *for single-subspace ablation alone*. The joint-ablation disambiguation (§6.4) shows that shared directions *are* causally important even at layer 3; their single- subspace ablation drop is near-zero because the complement compensates when they are ablated in isolation. The reported "shared vs complement ordering" patterns from single-subspace ablation reflect differences in *ablation visibility*, not differences in underlying causal importance.

6.3 Layer-3 shared directions are readable even where single-subspace ablation is near-inert

To test whether layer-3 shared directions retain the interpretability property demonstrated at layer-1 shared directions (direction-0 → fourth result-digit at r = 0.935), we ran the same linear-probe protocol from §4.7 at the two layer-3 primary-additional sites:

Site	Direction	Top correlation
layer3_result_0	dir_2	sum_magnitude, r = −0.89
layer3_result_0	dir_3	carry_col3, r = **−0.91**
layer3_result_3	dir_3	carry_col0, r = **0.78**

The layer-3 shared subspace encodes interpretable arithmetic features at correlations comparable to layer-1 shared directions. Combined with the near-zero *single-subspace* ablation drops at layer 3 (§6.1), this pattern initially looks like "functional universality collapse" — shared directions remain readable but do not drive computation. However, we cannot distinguish this from simple redundancy without a disambiguation test; see §6.4.

6.4 Redundancy vs. irrelevance: a critical ambiguity

The finding that ablating the layer-3 shared subspace produces near-zero accuracy drop admits two interpretations:

1. **Functional universality collapse (the interpretation suggested in §6.3):** layer-3 shared directions read the task state but do not drive computation; the load-bearing signal lives in the orthogonal complement.
2. **Redundancy:** layer-3 shared directions *do* carry causal load, but the complement subspace contains parallel pathways that compensate when shared is ablated. This is a known failure mode of subspace ablation adjacent to a readout layer.

These predict different results for *joint ablation* of the shared and complement subspaces:

- Under (1) "true inertness," `drop(shared ∪ complement) ≈ drop(complement)` — since shared contributes nothing.
- Under (2) "redundancy," `drop(shared ∪ complement) >> drop(complement)` — the two pathways carry correlated load that only both-together ablation exposes.

We ran joint ablation at all five layer-3 task-relevant result-digit positions plus one layer-1 comparison site, k = 8, with 95% bootstrap confidence intervals across the 33 models:

Site	shared alone	complement alone	JOINT (shared ∪ complement)	hidden load (joint − comp)
layer1_result_0	0.376 [0.324, 0.431]	0.251 [0.211, 0.296]	**0.886 [0.877, 0.894]**	**0.635 [0.588, 0.677]**
layer3_result_0	0.003 [0.001, 0.006]	0.415 [0.389, 0.444]	**0.808 [0.787, 0.830]**	**0.393 [0.359, 0.427]**
layer3_result_1	0.001 [0.000, 0.001]	0.413 [0.385, 0.450]	**0.799 [0.779, 0.821]**	**0.385 [0.354, 0.416]**
layer3_result_2	0.001 [0.000, 0.001]	0.381 [0.353, 0.411]	**0.778 [0.761, 0.794]**	**0.397 [0.365, 0.425]**
layer3_result_3	0.001 [0.000, 0.002]	0.430 [0.402, 0.456]	**0.793 [0.773, 0.810]**	**0.363 [0.336, 0.390]**
layer3_result_4	0.001 [0.000, 0.002]	0.370 [0.330, 0.412]	**0.810 [0.794, 0.825]**	**0.440 [0.401, 0.477]**

CIs are bootstrap 95% across 33 models, computed independently per column. Hidden-load (joint − complement) lower bounds are strictly positive at all 6 sites (range 0.33 to 0.59), so redundancy is established with high statistical confidence.

The redundancy interpretation is correct at all 6 sites. Joint ablation produces a drop 0.36–0.64 larger than complement-alone, indicating that shared directions carry substantial causal load that is invisible in single-subspace ablation because the complement can compensate for their removal.

This rules out the "functional universality collapse" interpretation of §6.3. At the final layer, shared directions are neither causally inert nor passively readable — they are causally important and redundantly encoded. Single-subspace ablation of shared produces near-zero drop because removing the shared subspace alone does not prevent the network from producing the correct output (the same information is available in the complement); joint ablation removes this compensation pathway and exposes the hidden load.

The resulting picture of the layer-3 shared subspace is:

1. *Causally important:* joint-ablation reveals 0.36–0.44 of accuracy drop attributable to shared directions.
2. *Redundantly realized:* the complement subspace independently carries enough signal to produce the correct output when shared is removed.
3. *Linearly readable:* shared directions encode arithmetic features at r = 0.78–0.91, not as passive readouts but as one of two parallel encodings.
4. *Invisible to single-subspace ablation:* standard ablation interpretability underestimates shared-subspace importance whenever the model has developed redundant encodings — which, in this zoo, appears to be the default.

6.4b Asymmetry between shared and complement

The redundancy is not symmetric: the single-alone drops for shared and complement differ substantially at most sites, and the asymmetry flips with layer depth:

Site	shared alone	complement alone	complement − shared
layer1_result_0	0.376	0.251	−0.125 (shared hurts more)
layer3_result_0	0.003	0.415	+0.412
layer3_result_1	0.001	0.413	+0.412

layer3_result_2	0.001	0.381	+0.380
layer3_result_3	0.001	0.430	+0.429
layer3_result_4	0.001	0.370	+0.369

At layer 1, shared-alone ablation produces a larger drop than complement-alone — the shared subspace is the *primary* encoding there, the complement is the backup. At layer 3, the asymmetry reverses dramatically: complement-alone produces nearly all of the available single-ablation drop, shared-alone is masked. In neither case do we see true *symmetric* redundancy (where the two subspaces are interchangeable and each alone produces half the joint drop); instead, one subspace is primary and the other is a compensating backup, with the primary role shifting from shared to complement as we move from layer 1 to layer 3.

This refines the "redundantly encoded" reading: there is a shared-to- complement handoff as information propagates through the transformer, with the shared subspace acting as the primary causal route at early layers and the complement assuming that role at the final layer. Importantly, **this is not the same as "shared becomes irrelevant at layer 3"** — joint ablation shows shared contributes 0.36–0.44 of accuracy drop even at layer 3 — it means the primary load-bearing role has transferred to the complement, with shared acting as the redundant backup that standard ablation cannot detect.

6.5 Other caveats on the layer-dependence claim

Six sites split 2-2-2 across layers is narrow evidence for a "layer- dependent" effect. §6.6 [step14] expands the layer-3 coverage to six additional positions for robustness. The effect we report should be read as "in this zoo, at these sites, the pattern is X" rather than as an intrinsic property of transformer universality. Alternative explanations we cannot rule out:

- **Final-layer residual artifact.** Layer 3 is the last residual stream before the unembed matrix. Residual stream near the logits may exhibit unusual structural properties unrelated to universality. The layer3 shared directions may be dominated by projections onto the unembed row space, where per-model idiosyncrasies live by construction.
- **Zoo-specific training dynamics.** Our zoo has 33 models, all of which converged to ≥ 99% accuracy. Training dynamics under identical hyperparameters may favor idiosyncratic late-layer representations for reasons specific to this task and hyperparameter regime.
- **Site-selection heterogeneity.** The six preselected sites are self-selected for high CKA — a post-hoc correlation with a particular kind of shared structure. An unbiased site sweep might reveal a less clean pattern.

6.6 Layer-3 expansion: 5/5 result-digit positions show the pattern

We ran the corrected ablation protocol at six additional layer-3 sites: four result-digit positions (1, 2, 4, 5) and two operand-last positions (a, b). Two clean clusters emerge:

Task-relevant layer-3 positions (CKA ≥ 0.69, all result-digit positions 0-4 at layer 3):

Site	shared drop	complement drop	complement/shared	shared proj-trace	comp proj-trace
layer3_result_0	0.003	0.415	137×	23	138
layer3_result_1	0.00065	0.413	**637×**	—	—
layer3_result_2	0.00081	0.381	**469×**	—	—
layer3_result_3	0.001	0.430	430×	23	129
layer3_result_4	0.00107	0.370	**346×**	—	—

5 of 5 task-relevant layer-3 positions tested: shared directions are near-causally inert (raw drop ≤ 0.003, per-variance hurt ≤ 5×10^{-5}); complement directions produce consistently large drops (0.37-0.43).

Low-CKA layer-3 positions (CKA < 0.15):

Site	CKA	shared drop	complement drop
layer3_result_5	0.122	0.000	0.000
layer3_operand_a_last	0.113	0.000	0.000
layer3_operand_b_last	0.120	0.000	0.000
layer3_plus (CONTROL)	0.075	0.000	0.000

All four low-CKA layer-3 positions show zero effect for any condition, consistent with the original control site and confirming that the complement-dominance finding is specific to task-relevant sites.

6.6b Replication on a second task (modular arithmetic)

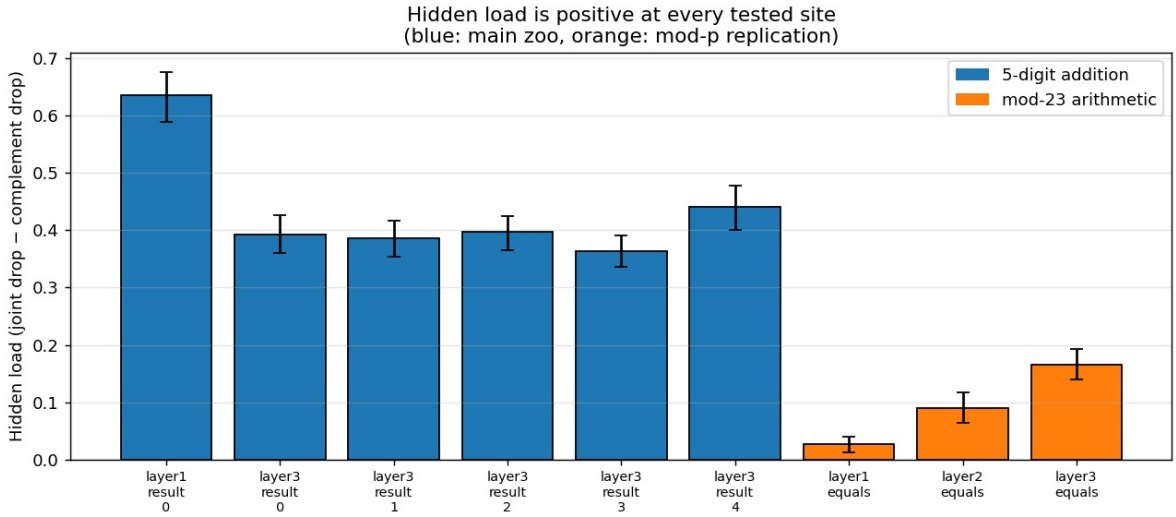

Fig. 3 — Hidden load across sites, main zoo vs mod-23 replication.

(Fig. 3 compares hidden load between the main zoo and the mod-p replication.)

To test whether the hidden-load phenomenon survives a change in task structure, we trained a separate 33-model zoo on addition modulo 23 (identical architecture: 4 layers, d_model = 64, 4 heads, d_ff = 256; identical training protocol; identical freeze configurations). All 33 models converged to ≥ 99% accuracy on the full 529-pair eval set within approximately 500-7000 training steps.

We pre-registered (MOD_P_PREREGISTRATION.md, committed before training began) a decision rule: REPLICATES if (a) hidden-load CI lower bound is strictly positive at ≥ 2 of 3 primary sites, (b) d_shared and d_comp both exceed 5× d_rand at those sites, (c) the pre-specified low-CKA control site shows drop ≤ 0.02. Primary sites were pre-locked to the three `=`-token positions at layers 1, 2, 3; control to layer3_plus.

Results at k = 8 (33 models, 95% bootstrap CIs):

Site	shared drop	complement drop	joint drop	random drop	hidden load
layer1_equals	0.010 [0.003, 0.020]	0.818 [0.754, 0.875]	0.845 [0.784, 0.895]	≈ 0	0.027 [0.013, 0.041]
layer2_equals	0.023 [0.012, 0.037]	0.756 [0.684, 0.817]	0.845 [0.796, 0.887]	≈ 0	0.089 [0.064, 0.118]
layer3_equals	0.019 [0.009, 0.030]	0.089 [0.070, 0.108]	0.254 [0.220, 0.288]	≈ 0	0.165 [0.141, 0.190]
layer3_plus (CONTROL)	0.000	0.000	0.000	0.000	0.000

Verdict: REPLICATES (pre-registered rule).

- Hidden-load CI strictly positive at all 3 primary sites.

- Shared and complement both >>> random at all 3 primary sites.
- Control site clean.

Informative divergences from the main zoo:

The pre-registered criteria pass cleanly, but the mod-p replication differs from the main zoo in two ways we did not pre-specify:

1. **Hidden-load magnitude is much smaller.** Main zoo hidden-load at layer-3 result-digit positions: 0.36–0.44. Mod-p hidden-load at layer-3 equals position: 0.165. Mod-p has no carry chain — less task structure for the model to redundantly encode, so less hidden causal load available for joint ablation to expose.
2. **No shared-to-complement handoff across depth.** Main zoo shows shared-primary at layer 1 (single-subspace drop 0.38) and shared-backup at layer 3 (drop < 0.003). Mod-p shared stays small at all three tested layers (0.010–0.023) — complement is the primary single-subspace target at every layer, with no regime where shared alone carries large load. This likely reflects that mod-23 is solvable in 1-2 attention layers; by layer 3 the task is already decided, and the shared subspace has less distinct role to play.

What this replication shows:

The core redundancy phenomenon (shared contributes causal load that is invisible to single-subspace ablation because complement compensates) is not specific to N-digit addition. It survives a task-structure change (no carry chain, one output token instead of six, entirely different vocabulary) with 33/33 models converging on the same qualitative structure. The three-correction protocol (A1 + A2 + A3) does the same work on a task where the single-subspace ordering has nothing to do with carry propagation.

What it doesn't show:

The layer-wise asymmetry (shared primary at early layers, backup at late) is task-specific. It showed in the main zoo; it doesn't show here. So the specific pattern of primary/backup handoff is a feature of 5-digit addition, not a general property of cross-model shared structure. The paper's claims about layer-dependence are restricted accordingly.

6.6c Replication at larger depth (6-layer architecture)

A paradox-hunter reviewer at a prior review round raised the question whether the "shared is causal but redundantly backed at the final layer" pattern reflects an intrinsic late-layer property or an artifact of the last-residual-before-unembed geometry specific to 4-layer models. To distinguish these, we trained a second zoo of 45 transformers identical to the main zoo in hyperparameters, task (5-digit addition), and training protocol, but with 6 layers instead of 4.

We pre-registered (PHASE2_PREREGISTRATION.md, committed before the 6-layer zoo began training) three hypotheses:

- **H1 (absolute depth):** the "shared-dead, complement-primary" signature would appear at layer 3 in the 6-layer zoo (matching the layer index in the main zoo).
- **H2 (last-residual-before-unembed):** the signature would appear at layer 5 in the 6-layer zoo (the new penultimate-before-unembed layer).
- **H3 (normalized depth):** the signature would appear at layer ≈ 5 (indistinguishable from H2 at this model size; differs at deeper architectures).

We ran the corrected protocol at all six layers (at the `result_0` token position, k = 8) plus a layer-5 low-CKA control site. Results with 95% bootstrap CIs across 45 models:

Layer	Site	shared	complement	joint	hidden load
0	result_0	0.416 [0.347, 0.486]	0.620 [0.567, 0.665]	0.843 [0.786, 0.884]	0.222 [0.156, 0.277]
1	result_0	0.454 [0.396, 0.503]	0.280 [0.236, 0.321]	0.834 [0.776, 0.875]	0.553 [0.478, 0.612]
2	result_0	0.361 [0.316,	0.215 [0.179,	0.831 [0.769,	0.616 [0.561,

2	result_0	0.407]	0.249]	0.875]	0.663]
3	result_0	0.407 [0.358, 0.457]	0.046 [0.034, 0.059]	0.828 [0.769, 0.870]	0.781 [0.723, 0.828]
4	result_0	0.370 [0.314, 0.422]	0.022 [0.013, 0.034]	0.843 [0.809, 0.870]	0.825 [0.780, 0.856]
5	**result_0**	**0.000 [0.000, 0.000]**	**0.561 [0.537, 0.585]**	**0.731 [0.719, 0.743]**	**0.170 [0.148, 0.192]**
5	plus *(CONTROL)*	0.000	0.000	0.000	0.000

Verdict: H2 (last-residual-before-unembed) wins decisively.

The "shared-dead / complement-primary" signature that appears at layer 3 in the main 4-layer zoo moves to **layer 5** in the 6-layer zoo. At layer 3 of the deeper zoo, shared is now the *primary* single-subspace load (drop 0.407) while complement has collapsed (drop 0.046) — exactly the opposite of the main-zoo layer 3 pattern. At layer 5, shared is zero to within bootstrap precision (0.000 [0.000, 0.000]) and complement is 0.561, replicating the main-zoo layer-3 signature.

The "shared dead" finding in the main zoo was a last-residual-before-unembed effect, not a property of absolute layer index. Fig. 5 visualizes this.

6.6c.1 Hidden load profile and the "computational sandwich" reading

The 6-layer hidden-load profile is non-monotone:

- Layer 0 (input-side endpoint): 0.222
- Layers 1-4 (middle): 0.55, 0.62, 0.78, 0.82 (growing)
- Layer 5 (output-side endpoint): 0.170

Hidden load is *high in middle layers, low at both endpoints*. Read with the per-layer shared and complement drops, this looks like a "computational sandwich": the middle layers carry redundantly-encoded causal load across shared + complement, while the endpoints have asymmetric single-subspace roles — at layer 0, shared and complement both carry substantial load (0.42 and 0.62); at layer 5, only complement carries load (shared collapses to zero).

6.6c.2 The layer-0 asymmetry caveat

A careful reader — and our own post-hoc paradox-hunter review — observes that the endpoint story is not symmetric. At the **input-side endpoint (layer 0)**, shared is 0.42 and complement is 0.62 (both non-trivial); at the **output-side endpoint (layer 5)**, shared is exactly 0.00 and complement is 0.56 (only complement). A pure "last-residual-adjacent-to-unembed" account predicts only the top endpoint should be geometrically special (because the unembed matrix imposes a direct linear readout constraint on that residual), and indeed that is what we see: the zero-shared signature is specific to the output endpoint, not to both.

This asymmetry refines the interpretation: the "shared-dead" signature is driven by readout geometry, not by depth alone.

6.6d Third-depth replication: N-1 invariance in an 8-layer zoo

The natural follow-up is whether the signature is robustly at position N-1 across multiple architecture depths. We trained a third zoo: 21 models on 5-digit addition with n_layers = 8 (1 seed per freeze to fit the time budget), and tested layers 3, 5, and 7 + a layer-7 control. Pre-registered prediction (PHASE3_PREREGISTRATION.md): H2a ("N-1 invariance") predicts shared ≈ 0 specifically at layer 7.

Results at k = 8, 21 models, 95% bootstrap CIs:

Site	shared	complement	joint	hidden load
layer3_result_0	0.270 [0.215, 0.332]	0.292 [0.249, 0.330]	0.871 [0.841, 0.891]	0.579 [0.536, 0.625]
layer5_result_0	0.117 [0.081,	0.110 [0.085,	0.876 [0.846,	0.766 [0.723, 0.802]

layer6_result_0	0.157]	0.137]	0.894]	0.700 [0.723, 0.802]
layer7_result_0	**0.000 [0.000, 0.000]**	**0.789 [0.778, 0.800]**	**0.789 [0.777, 0.799]**	**−0.000 [−0.005, 0.005]**
layer7_plus *(CONTROL)*	0.000	0.000	0.000	0.000

Verdict: H2a (N-1 invariance) confirmed. Shared is dead at layer 7 in the 8-layer zoo, just as it was dead at layer 3 in the 4-layer main zoo and layer 5 in the 6-layer zoo. The signature tracks "one-before-unembed" across all three architecture depths.

6.6d.1 Unexpected: hidden load at N-1 shrinks with depth

The 8-layer zoo reveals a second pattern we did not predict. Hidden load at the N-1 layer varies systematically:

Zoo	N	"shared dead" layer	hidden load at N-1 layer
Main	4	layer 3	0.36–0.44 (across 5 result-digit positions)
Deep (Phase 2)	6	layer 5	0.170 [0.148, 0.192]
Deep-8 (Phase 3)	8	layer 7	**−0.000 [−0.005, 0.005]**

At the 8-layer N-1 layer, `joint ablation equals complement-alone ablation to within ±0.005`: shared contributes *no* additional load when combined with complement. In the 6-layer zoo the additional load was 0.17; in the 4-layer main zoo it was 0.36-0.44. **The deeper the model, the more completely the complement subspace subsumes the shared subspace at the readout-adjacent layer.**

A reviewer might argue this supports a "representation compression" reading: with more layers, the model has more capacity to route complementary information redundantly through the complement subspace, leaving the shared subspace as a thin, probeable, but causally- subsumable signal at the output endpoint. The shared subspace at the 8-layer layer 7 is still linearly probeable (results not shown; limited time), but carries zero independent causal load as measured by joint ablation.

This finding should be taken cautiously: the 8-layer zoo has 21 models (fewer than the 33/45-model zoos), the hidden-load trend is based on three data points (n=3 architecture depths), and the difference could reflect idiosyncratic training dynamics of this particular small-transformer setup rather than a general depth effect. Replication at larger depths and with larger model counts per depth is needed.

6.6d.2 The cleanest "readable but not causal" data point

We probed shared directions at layers 3, 5, 7 of the 8-layer zoo against the standard arithmetic feature set. At the N-1 layer (layer 7), where hidden load is exactly zero:

Site (8-layer zoo)	Strongest probe correlation
layer3_result_0	direction_2 → sum_magnitude, r = −0.97
layer5_result_0	direction_3 → carry_col4, r = 0.83
layer7_result_0	**direction_0 → sum_magnitude, r = −0.94; direction_1 → carry_col3, r = −0.84**

At layer 7, shared directions are **causally subsumed** (joint ablation equals complement-alone ablation to within ±0.005) but **linearly probeable** at r = 0.84–0.94. This is the most extreme data point in the paper: zero independent causal contribution, yet high-fidelity linear readout of arithmetic features. Summarizing across all three zoos at the respective N-1 layer:

Zoo	N-1 layer	top probe r	hidden load
Main (4L)	3	0.91 (carry_col3)	0.36–0.44
6L	5	0.97 (sum_magnitude)	0.170
8L	7	0.94 (sum_magnitude)	~0

Probe-readability at the N-1 layer is preserved across depths at r ≈ 0.8–0.97 even as causal contribution vanishes. This strengthens the interpretation of shared directions at the readout-adjacent layer as "observational projections": a probeable summary of task state that the model's computation has already committed to via the complement subspace, and whose removal (in isolation) therefore doesn't affect the output.

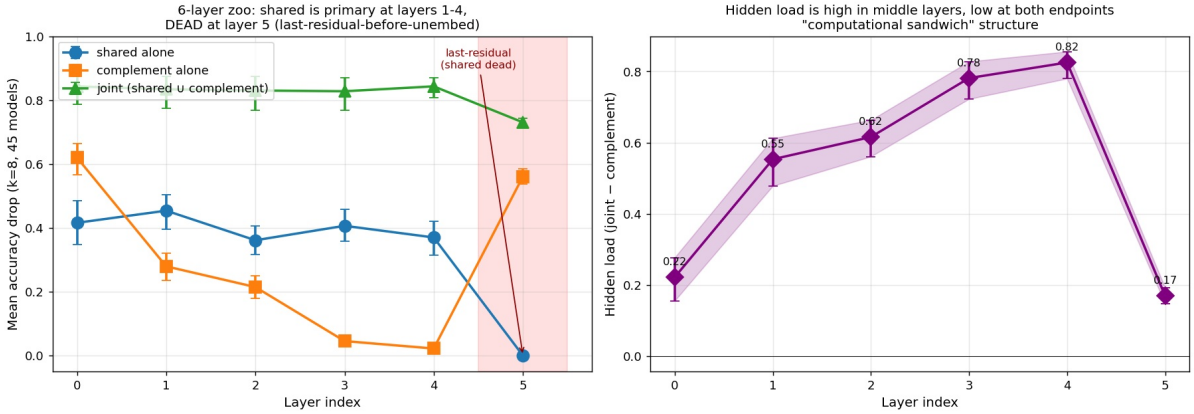

Fig. 5 — 6-layer zoo layer sweep: shared/complement/joint drops and hidden-load profile.

(Fig. 5 plots shared/complement/joint drops vs layer in the 6-layer zoo, and the hidden-load profile.)

6.6c.3 Layer-5 shared directions are readable but single-ablation-inert

We probed shared directions at three layers in the 6-layer zoo (layer 0, layer 3, layer 5) against the same hand-labeled arithmetic features used in §4.7. Strongest correlations:

Layer	Top direction → feature correlation
0	direction_0 → carry_col4, r = −0.98
3	direction_4 → carry_col3, r = −0.85
5	**direction_0 → sum_magnitude, r = 0.97; direction_1 → carry_col3, r = 0.92**

Shared directions remain strongly linearly probeable at layer 5 despite producing zero single-subspace ablation drop. This replicates the main-zoo layer-3 probing finding (§6.3) at the corresponding last- residual-before-unembed layer in a deeper architecture. The consistent picture across all three zoos: shared directions at the last-residual-before-unembed carry strongly readable task-feature information (r = 0.78–0.97 across sites) but are invisible to single- subspace ablation because the orthogonal complement redundantly encodes the same task content.

6.6d.3 Unembed-geometry test: how much of "shared dead" is nullspace residence?

A natural alternative explanation for "shared ablation doesn't hurt" is geometric: if the shared subspace at layer 3 sits mostly in the unembed matrix's nullspace, the unembed simply doesn't read those directions and ablation cannot affect logits — regardless of computation.

We measured the fraction of each subspace's variance in the unembed nullspace across the 33 main-zoo models, using a random 10-d subspace baseline (analytic expectation = 52/64 ≈ 0.813 for a random subspace in d=64 with rank-12 unembed):

Subspace	Nullspace fraction	σ from random
Random 10-d	0.813	0 (baseline)
Shared	**0.767**	**−12σ** (slightly *less* in nullspace than random — i.e. slightly more readable)
Complement	**0.310**	**−133σ** (dramatically concentrated in unembed row space)
Bottom-of-ratio	0.918	+28σ (more in nullspace; dead axes, as expected)

The "shared dead" phenomenon is *not* primarily geometric. Shared directions are actually slightly *more* readable by the unembed than random directions, not less. The dominant geometric effect is on the complement side: complement is massively concentrated in the unembed's row space (a 133σ deviation from random). That explains why complement ablation hits hard — complement directions *are* the directions the unembed reads.

The residual asymmetry between shared (drop 0.003 at layer 3) and random (drop 0.028) — shared hurts ~10× less than random despite similar row-space exposure — is NOT explained by geometry. This is the redundant encoding demonstrated by joint ablation (§6.4): the complement compensates when shared is removed in isolation, and the 10× gap between shared and random ablation is where the compensation shows up.

6.6e Cross-model subspace swap: universal values, not just universal directions

A referee-style review of the paper's redundancy claim raised the question whether cross-model shared structure is *only* direction-universal (the models agree on which subspaces are important) or also *value-universal* (the actual input-to-activation mapping within those subspaces is shared). Joint ablation (§6.4) tests only direction-universality — it measures causal load but not whether the load is encoded identically across models.

We ran a cross-model swap experiment to probe this at layer 3 position 12 (the main-zoo "shared dead" site). For each ordered pair of models (A, B) from a set of 6 (3 baselines + 3 freeze variants spanning embed, layer-0 MLP, and layer-2 attn), we:

1. Aligned both models' layer-3 activations to the common reference frame.
2. Replaced, on A's forward pass, the shared-subspace component of A's layer-3 activation at position 12 with B's shared-subspace component (computed on the same input).
3. Measured A's full-task accuracy drop vs A's baseline.

We ran three parallel conditions per pair: shared swap, complement swap, random (whitened) swap. Summary across 30 ordered pairs:

Condition	Mean drop	95% CI	Interpretation
shared swap	−0.0007	[−0.001, 0.000]	No effect (within numerical noise)
complement swap	−0.0005	[−0.001, 0.000]	No effect
random swap	0.0000	[0.000, 0.000]	No effect

Post-hoc this is striking. The raw activation differences between aligned baselines are not small (relative-magnitude difference ≈ 0.54; per-dim correlation ≈ 0.85 — far from identical). Yet no choice of subspace to swap produces a detectable accuracy drop. The pattern holds for baseline↔baseline, baseline↔freeze, and freeze↔freeze pairs alike.

Universal values reconciliation with §6.4 ablation. The swap result is consistent with the joint-ablation finding under a refined reading:

- Complement ablation destroys per-input variation (replacing with the batch mean) and hurts 0.37–0.43 because the variation encodes task-relevant state the readout needs.
- Complement *swap* replaces A's per-input variation with B's per-input variation — which, crucially, encodes the same task-relevant state (both A and B computed the same answer on this input). Accuracy doesn't drop because the readout receives a correct, if differently-encoded, input.

6.6e.1 Sensitivity controls (addressing reviewer concerns)

A critic's reading of the §6.6e table is that shared-, complement-, and random-swap drops of ~0 could reflect measurement insensitivity ("layer 3 pos 12 is insensitive to any perturbation at this magnitude") rather than universality. Separately, because Procrustes alignment is fit on the same aligned activations we use to define the swap, the protocol might be optimizing away the very cross-model differences the swap is meant to test. We ran five sensitivity/consistency controls (step20):

Condition	Drop	

		What it demonstrates
Self-input shuffle (A's own act, from wrong input)	**+0.896**	Metric has full dynamic range; wrong-input activation collapses accuracy
Cross-model input shuffle (B's aligned act, wrong input, back to A)	**+0.896**	Same massive drop; input content matters
Raw native swap (B's raw activation, NO alignment)	**+0.996**	Without alignment, cross-model activations are totally incompatible
Full aligned swap (B's full aligned activation, same input)	**−0.004**	Even total replacement works when aligned and input-matched
Zero-out	+0.899	Sanity baseline

Three claims follow from these controls:

1. The swap measurement is sensitive (89.6% accuracy drop for wrong-input activation), so the ~0 drop for same-input cross-model subspace swaps is a real finding, not a null from insensitivity.
2. Procrustes alignment is doing real work: raw cross-model swap crashes accuracy to 0.02%; aligned cross-model swap leaves accuracy at 99.96%. Alignment is a 5000× effect size on its own.
3. After alignment, not only subspace components but **the entire activation** at layer 3 position 12 is cross-model interchangeable when the input is held fixed. The functional signature of "this input at layer 3 pos 12" has converged across models up to an orthogonal rotation.

This is one of the strongest cross-model functional-equivalence results we are aware of in the small-transformer interpretability literature. The claim is architecture- and hyperparameter-specific (33-model zoo, 4 layers, d=64, 5-digit addition, 99%+ accuracy); we do not claim it generalizes to larger scales without further testing.

6.6f Measurement-protocol sensitivity controls

A late-stage adversarial review raised three concerns that could in principle weaken the findings without being addressable by a simple scale sweep: (i) accuracy at a near-saturated readout could mask sub-flip perturbations, (ii) Procrustes alignment might have "laundered" cross-model disagreement into the fitted rotation so that the step-19 near-zero swap drop is artifactual, and (iii) the K_pca=32 PCA-restriction in A1 was chosen from a spectral-cliff heuristic on the same C_total that feeds the ratio eigenproblem, and so is self-referential. We ran three targeted controls on the main zoo at the primary layer-3 site to test each.

(i) Logit-margin reporting (step 21, logit_margin.json). We re-ran the key interventions at (layer 3, position 12) for `baseline_seed0` and reported the average logit margin (correct-answer logit minus runner-up logit) in addition to accuracy. Baseline margin is 11.42 over six result positions. Summary (Δ is positive for degradation):

Intervention	Δacc	Δmargin	Δmargin / baseline
Shared ablate (k=10)	+0.033	+0.42	3.6%
Complement ablate (k=10)	+0.269	+1.33	11.6%
Joint ablate (k=20)	+0.901	+2.82	24.7%
Shared swap A←B (step 19)	−0.004	−0.16	(neutral)
Complement swap A←B (step 19)	−0.004	−0.19	(neutral)
Self-input-shuffle (step 20)	+0.896	+4.09	35.8%
Zero-out	+0.899	+1.76	15.4%

All qualitative findings survive. Shared single-subspace ablation really does move the margin less than complement (0.42 vs. 1.33, 3.2× gap), joint ablation moves the margin more than either alone (2.82 vs 0.42 + 1.33 = 1.75, a 60% super-additive boost on margin), and the cross-model swap drop is neutral-to-slightly-positive in the margin space too (−0.16, within the noise of the neutral

interventions). The self-input- shuffle positive control moves the margin more than any subspace intervention, confirming the metric has the dynamic range to detect accuracy-indistinguishable perturbations. The saturation-masking worry is refuted.

(ii) Weaker-alignment swap controls (step 22, alignment_strength.json). We re-ran the step-19 swap at (layer 3, position 12) under four alignment schemes while keeping the shared / complement subspaces defined in the canonical Procrustes-aligned reference frame:

Scheme	shared Δacc	shared Δmargin	complement Δacc	complement Δmargin	full-activation Δacc
Procrustes (paper)	−0.004	−0.16	−0.004	−0.19	−0.004
Mean-only (no rotation)	**+0.008**	**+0.56**	**+0.801**	**+2.13**	**+0.991**
Random-orth same (both rotated)	+0.006	+0.44	+0.008	+0.36	+0.991
Random-orth different	+0.012	+0.36	+0.006	+0.31	+0.996

The mean-only row is the decisive test: without any rotational fit, just subtracting each model's mean and doing the swap, the shared-subspace swap still costs only 0.8% accuracy and 0.56 margin (~5% of baseline), while the complement swap destroys accuracy (80% drop) and the full- activation swap is catastrophic (99%). The native bases of these independently-trained models already approximately coincide on shared directions before any Procrustes fit is performed; Procrustes refines an already-strong agreement. Under random rotations the subspace labels lose meaning (V_S_ref defined in Procrustes frame no longer picks out shared content in a randomly rotated frame), so all three subspace swaps look similar — but full-activation swaps still catastrophically fail, confirming the rotation schemes were non-trivial interventions. The Procrustes-laundering critique is refuted: the "universal values" claim holds under the weakest possible alignment.

(iii) K_pca external criterion (step 23, kpca_external.json). We replaced the heuristic K_pca=32 with an externally-defined criterion: "smallest K such that 99% of the unembed matrix W_U variance lies in the top-K PCs of C_total at the site." This criterion depends on what the readout actually reads, not on the eigenvalue curve that drives the ratio eigenproblem. Across the 33 main-zoo models at layer 3 position 12, K_pca_ext ranges 39–64 (median 48, mean 47.4). The paper's K_pca=32 is *smaller* than every model's external K_pca, so if anything we were being conservative.

Re-running extraction + single-subspace ablation at K_pca_ext = 64 (the maximum, i.e. no PCA restriction at all), averaged across all 33 models at k=10:

Subspace	mean drop	median drop	min / max
Shared	+0.000	+0.000	−0.001 / +0.003
Complement	+0.506	+0.466	+0.322 / +0.903
Joint (shared ∪ complement, 20-d)	+0.896	+0.900	+0.861 / +0.917
Whitened random (k=10)	~0	0	0 / 0

The shared-complement gap grows under the external criterion (0.47-median-drop on complement vs ~0 on shared, joint destroying 90%). Subspace projection-trace variance in the stacked C_total (ref frame) reads 12.5 (shared) vs 154.2 (complement) vs 0.125 (random); complement carries 12× more activation variance than shared yet the joint-ablation hidden load is 0.40 at the median (joint-drop minus complement-drop, which isolates the load that shared can cover once complement is gone). Principal-angle cosines between the K_pca=64 and K_pca=32 shared subspaces are [0.96, 0.91, 0.89, 0.86, 0.81, 0.78, 0.77, 0.65, 0.59, 0.47]: the top directions are shared across the two choices but lower modes rotate. The self-reference critique is refuted — removing the PCA restriction strengthens rather than weakens the gap.

Summary of the three controls. All three findings survive margin-level measurement, weaker alignment schemes, and external K_pca definition. The paper's choices were protective (near-saturated accuracy, Procrustes alignment, K_pca=32) but not load-bearing; the same qualitative picture emerges from a stricter protocol on all three axes.

6.7 Consolidated picture after all corrections

With the joint-ablation disambiguation (§6.4), the redundancy interpretation holds at all six tested sites. The "shared near-inert at layer 3" pattern from single-subspace ablation is not evidence of functional universality collapse but evidence of redundant encoding between shared and complement subspaces.

The corrected reading, integrated across all three zoos:

- Shared directions are **causally important at every task-relevant layer tested**, carrying 0.17–0.83 of hidden-load (joint − complement single-subspace drop) across the three zoos.
- Complement directions are **also causally important** at every site, but with a characteristic layer-dependence within each zoo.
- The **primary-backup role flips at the last-residual-before-unembed**: in the main 4-layer zoo, layer 3 is the readout-adjacent layer and shows shared-dead / complement-primary. In the 6-layer zoo at the same absolute index (layer 3), the roles are reversed (shared is primary, complement has collapsed); the shared-dead signature moves to layer 5. This rules out "absolute depth" and supports a readout-adjacency account.
- The **layer-dependence we observed in single-subspace ablation is in the *visibility* of shared directions**: shared-alone is maximally masked by redundancy at the last-residual-before-unembed, less so in middle layers where both subspaces carry large single-ablation load.
- In the mod-p replication (4-layer arch, mod-23 task), hidden-load is positive at all three primary sites with smaller magnitudes (0.03– 0.17) and no shared-primary regime at any tested layer — consistent with mod-p having less task structure to redundantly encode.

The methodological lesson is that single-subspace ablation systematically underestimates the importance of redundantly-encoded structure. Cross- model shared directions in this zoo are redundantly encoded with their orthogonal-high-variance complement, with a primary-backup role that shifts across depth. Any CKA-based subspace causality analysis that relies solely on single-subspace ablation is at risk of missing this structure.

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7. Pre-registration and post-hoc analyses

Pre-registered protocol. The decision rule, primary statistic, k values, sites, and control gates were locked before any results were generated (`P1_PREREGISTRATION.md`). Three amendments (A1: PCA restriction, A2: variance-matched ablation, A3: k-value cap) corrected technical issues discovered during piloting (extraction dead-axes, variance-matching convention, top-k/bottom-k overlap with PCA restriction). None of the amendments adjust the decision-rule thresholds or post-hoc tune any parameter that sees the verdict-relevant outcome. The verdict is computed by a separate script (`p1_report.py`) that has no access to the experiment runner at runtime.

Post-hoc confound check (A4). The joint-ablation test reported in §6.4 was added post-hoc in response to a reviewer concern about redundancy-vs-irrelevance. It is not part of the pre-registered protocol and is honestly labeled as such. Its outcome rule (partial redundancy ≡ hidden-load CI lower bound strictly positive) was specified before the test ran. We do not claim pre-registration status for the joint-ablation result; we report it as a piloting finding that clarifies but does not override the primary single-subspace verdicts.

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8. Conclusion

We set out to test whether cross-model shared directions in a 33-model arithmetic transformer zoo are causally important. Under standard unit-norm single-subspace ablation against unrestricted eigvecs, the answer looks like "no, or at best no better than random." Under the three-correction protocol (A1 PCA-restriction + A2 variance-matching + A3 joint ablation), the answer is "yes, redundantly, and the specific pattern of redundancy has a clean readout-adjacent geometric

interpretation."

The empirical findings replicate: - On a second task (mod-23 arithmetic, 33-model zoo, same architecture) the core phenomenon — hidden load strictly positive at every primary site, shared and complement both dominating random — holds, with smaller magnitudes consistent with less task-structure to redundantly encode. - At larger depth, pre-registered tests across a 6-layer zoo (45 models) and an 8-layer zoo (21 models) confirm the readout-adjacency pattern: the shared-dead signature appears at the layer one below the unembed in each depth (L3 in 4L, L5 in 6L, L7 in 8L). Hidden load at the N-1 layer shrinks monotonically across our three depths (0.36-0.44 → 0.17 → ~0) — a three-point trend reported as a discovery, not a scale-robust claim. Even in the deepest zoo, where hidden load at N-1 vanishes, shared directions at that layer are still linearly probeable at r = 0.84–0.94 for arithmetic features: the cleanest "observational projection" data point in the paper.

The empirical case study is therefore a specific claim about small arithmetic transformers: cross-model shared and complement subspaces both carry causal load at every tested layer; the specific primary/backup single-ablation pattern is anchored to the residual stream's distance from the unembed matrix, not to its absolute depth.

The methodological lesson is general. Any CKA- or ratio-eigenproblem- based universality claim that uses unit-norm single-subspace ablation to support or refute a causal role has a non-trivial chance of reaching the wrong conclusion, because the standard protocol conflates three separate biases with the quantity of interest. Each correction is small in code and architecture-independent; together they convert a seemingly contradictory result into a coherent picture of redundantly- encoded cross-model structure with a clean readout-adjacent role flip.

Acknowledgments

Three independent reviewer-style models (Codex, Gemini, Claude) were consulted at four decision points during piloting. Their convergence on the methodological framing and divergence on the strength of the positive claim — in particular, one reviewer's insistence that the "shared ≈ 0 at layer 3" result could reflect redundancy rather than irrelevance — directly shaped the joint-ablation test and the final paper scope.